

$$\textcircled{1} f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}xy^2 + 2xy & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

First let's find z

$$\int_0^1 \int_0^1 \frac{1}{2}xy^2 + 2xy = 1$$

$$\int_0^1 xy^2 + 2xy \, dy = \left[\frac{1}{3}xy^3 + xy \right]_0^1 = \frac{1}{3}x + x = \frac{4}{3}x$$

$$\int_0^1 \frac{4}{3}x \, dx = \left[\frac{2}{3}x^2 \right]_0^1 = \frac{2}{3}$$

$$\frac{1}{2} \cdot \frac{2}{3} = 1$$

$$z = \frac{2}{3}$$

$$\textcircled{1.1} P[X+Y < 0.5]$$

$$A(x,y): x > 0, y > 0, x+y < 0.5$$

$$P[X+Y < 0.5] = \int_0^{0.5} \int_0^{0.5-x} \frac{1}{2}xy^2 + 2xy \, dy \, dx$$

$$\begin{aligned} \int_0^{0.5-x} \frac{1}{2}(xy^2 + 2xy) \, dy &= \frac{1}{2} \left[\frac{1}{3}xy^3 + xy^2 \right]_0^{0.5-x} = \frac{1}{2}x(0.5-x)^2 \left(\frac{1}{3}x(0.5-x) + 1 \right) \\ &= \frac{1}{2}x(0.5-x)^2(3.5x) = \frac{1}{2}(x(0.25x + x^2)(3.5x)) \\ &= \frac{1}{2}[0.875x - 3.75x^2 + 4.5x^3 - x^4] \end{aligned}$$

$$\begin{aligned} P[X+Y < 0.5] &= \int_0^{0.5} \frac{1}{2}[0.875x - 3.75x^2 + 4.5x^3 - x^4] \, dx \\ &= \frac{1}{2} \left[\frac{7}{16}x^2 - \frac{3.75}{3}x^3 + 2.25x^4 - \frac{1}{5}x^5 \right]_0^{0.5} \end{aligned}$$

$$P[X+Y < 0.5] \approx 0.00889$$

$$(1.2) \quad E[X] = \int x f_X(x) dx$$

$$f_X(x) = \int f_{X,Y}(x,y) dy = \int_0^1 \frac{3}{2} (xy^2 + 2xy) dy = \frac{3}{2} \left[\frac{1}{3} xy^3 + xy^2 \right]_0^1 \\ = \frac{3}{2} \left[\frac{1}{3} x + x \right] = 2x$$

$$E[X] = \int_0^1 x \cdot 2x dx = \int_0^1 2x^2 dx = \frac{2}{3} x^3 \Big|_0^1 = \frac{2}{3}$$

$$(1.3) \quad \text{Var}[Y] = E[Y^2] - E[Y]^2$$

$$E[Y] = \int y f_Y(y) dy$$

$$f_Y(y) = \int f_{X,Y}(x,y) dx = \int_0^1 \frac{3}{2} (xy^2 + 2xy) dx = \frac{3}{2} \int_0^1 x (y^2 + 2y) dx \\ = \frac{3}{4} x^2 (y^2 + 2y) \Big|_0^1 = \frac{3}{4} (y^2 + 2y)$$

$$E[Y] = \int y f_Y(y) dy = \int y \frac{3}{4} (y^2 + 2y) dy = \frac{3}{4} \int_0^1 y^3 + 2y^2 dy = \\ = \frac{3}{4} \left[\frac{1}{4} y^4 + \frac{2}{3} y^3 \right]_0^1 = \frac{3}{4} \left(\frac{1}{4} + \frac{2}{3} \right) = \frac{3}{4} \cdot \frac{11}{12} = \frac{11}{16} \\ = \frac{11}{16}$$

$$E[Y^2] = \int y^2 f_Y(y) dy = \int_0^1 y^2 \frac{3}{4} (y^2 + 2y) dy = \frac{3}{4} \int_0^1 y^4 + 2y^3 dy \\ = \frac{3}{4} \left[\frac{1}{5} y^5 + \frac{1}{2} y^4 \right]_0^1 = \frac{3}{4} \left(\frac{1}{5} + \frac{1}{2} \right) \\ = \frac{21}{40}$$

$$\text{Var}[Y] = E[Y^2] - E[Y]^2 \\ = \frac{21}{40} - \left(\frac{11}{16} \right)^2 = \frac{67}{1280}$$

$$(1.4) \operatorname{Cov}[X, Y] = E[XY] - E[X]E[Y]$$

$$E[XY] = \int \int xy \cdot f_{x,y}(x,y) dx dy$$

$$= \int_0^1 \int_0^1 xy \cdot \frac{3}{2}(xy^2 + 2xy) dy dx = \frac{3}{2} \int_0^1 \int_0^1 (x^2 y^3 + 2x^2 y^2) dy dx$$

$$\int_0^1 (x^2 y^3 + 2x^2 y^2) dy = x^2 \left(\frac{1}{4} y^4 + \frac{2y^3}{3} \right) \Big|_0^1 = x^2 \cdot \frac{11}{12}$$

$$E[XY] = \frac{3}{2} \int_0^1 \frac{11}{12} x^2 dx = \frac{3}{2} \cdot \frac{11}{12} \left(\frac{1}{3} x^3 \right) \Big|_0^1$$

$$= \frac{11}{24}$$

$$\operatorname{Cov}[X, Y] = E[XY] - E[X] \cdot E[Y] = \frac{11}{24} - \frac{2}{3} \cdot \frac{11}{16} = \frac{11}{24} - \frac{11}{24}$$

$$\operatorname{Cov}[X, Y] = 0$$

$$(1.5) \operatorname{Corr}[X, Y] = \frac{\operatorname{Cov}[X, Y]}{\sqrt{\operatorname{Var}[X] \cdot \operatorname{Var}[Y]}} = 0$$

$$(1.6) P[X > 0.5 | Y > 0.5] = \frac{P[X > 0.5 \cap Y > 0.5]}{P[Y > 0.5]}$$

$$P[X > 0.5 \cap Y > 0.5] = \int_{0.5}^1 \int_{0.5}^1 \frac{3}{2}(xy^2 + 2xy) dy dx$$

$$\int_{0.5}^1 \frac{3}{2}(xy^2 + 2xy) dy = \frac{3}{2} \left[\frac{1}{3} xy^3 + xy^2 \right] \Big|_{0.5}^1 = \frac{3}{2} \left[\left(\frac{x}{3} + x \right) - \left(\frac{x}{24} + \frac{x}{4} \right) \right]$$

$$= \frac{25x}{16}$$

$$P[X > 0.5 \cap Y > 0.5] = \int_{0.5}^1 \frac{25x}{16} dx = \frac{25}{16} \left(\frac{1}{2} x^2 \right) \Big|_{0.5}^1 = \frac{25}{16} \cdot \frac{3}{8}$$

$$= \frac{75}{128}$$

$$P[Y > 0.5] = \int_{0.5}^1 \frac{3}{4}(y^2 + 2y) dy = \frac{25}{32}$$

$$P\left[\frac{X \geq 0.5 \cap Y > 0.5}{Y \geq 0.5}\right] = \frac{3}{4}$$

$$(17) E[X | Y = 0.25]$$

$$\textcircled{3} U \sim U(0,1)$$

$$X \sim U$$

$$E[X] = \int_0^1 x f_X(x) dx = \int_0^1 x dx = \frac{1}{2} x^2 \Big|_0^1 = \frac{1}{2}$$

$$\text{Var}[X] = E[X^2] - E[X]^2 = \frac{1}{3} - \frac{1}{2} \cdot \frac{1}{2} = \frac{4}{12} - \frac{3}{12} = \frac{1}{12}$$

$$E[X^2] = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$

$$(4.1) \int_0^{\infty} x e^{-x} dx$$

$$u = \frac{x}{1+x} \quad x = \frac{u}{1-u} \quad dx = \frac{1}{(1-u)^2} du$$

$$u(1+x) = x$$

$$u + ux = x$$

$$u = x - ux$$

$$u = x(1-u)$$

$$\begin{aligned} \int_0^{\infty} x e^{-x} dx &= \int_0^1 \frac{u}{1-u} e^{-\left(\frac{u}{1-u}\right)} \frac{1}{(1-u)^2} du = \int_0^1 \frac{u}{(1-u)^3} e^{-\left(\frac{u}{1-u}\right)} du \\ &= \mathbb{E}\left[\frac{u}{(1-u)^3} e^{-\left(\frac{u}{1-u}\right)}\right] \end{aligned}$$

$$(4.2) u = 1 - e^{-x}$$

$$\frac{du}{dx} = e^{-x} \quad dx = \frac{1}{e^{-x}} du$$

$$e^{-x} = 1-u$$

$$dx = \frac{1}{e^{\ln(1-u)}} du$$

$$dx = \frac{1}{1-u} du$$

$$x = -\ln(1-u)$$

$$\begin{aligned} \int_0^{\infty} x e^{-x} dx &= \int_0^1 -\ln(1-u) e^{\ln(1-u)} \frac{1}{1-u} du = \int_0^1 -\ln(1-u) (1-u) \frac{1}{1-u} du \\ &= \int_0^1 -\ln(1-u) du \end{aligned}$$

$$= \mathbb{E}[-\ln(1-u)]$$

$$\textcircled{5} \quad X = \max\{U_1, U_2\}$$

$$\textcircled{5.1} \quad \theta = \mathbb{P}[X \leq 0.5]$$